

South Wall 16'-0" (192") – No door – West to East

Plates

PIECE	SIZE	LENGTH	NOTES
Bottom plate	2x6	16'-0"	
Lower top plate	2x6	16'-0"	
Upper top plate	2x6	15'-1"	Shortened 5-1/2" each end (end truss overlap)

Stud Layout (13 studs)

#	POSITION	TYPE	LENGTH
1	0 – 1-1/2"	Stud	6'-10-3/4"
2	15-1/4 – 16-3/4"	Stud	6'-10-3/4"
3	31-1/4 – 32-3/4"	Stud	6'-10-3/4"
4	47-1/4 – 48-3/4"	Stud	6'-10-3/4"
5	63-1/4 – 64-3/4"	Stud	6'-10-3/4"
6	79-1/4 – 80-3/4"	Stud	6'-10-3/4"
7	95-1/4 – 96-3/4"	Stud	6'-10-3/4"
8	111-1/4 – 112-3/4"	Stud	6'-10-3/4"
9	127-1/4 – 128-3/4"	Stud	6'-10-3/4"
10	143-1/4 – 144-3/4"	Stud	6'-10-3/4"
11	159-1/4 – 160-3/4"	Stud	6'-10-3/4"
12	175-1/4 – 176-3/4"	Stud	6'-10-3/4"
13	190-1/2 – 192"	Stud	6'-10-3/4"

All 13 studs identical: Cut 13 x 2x6 @ 6'-10-3/4" (82.75")

North Wall 16'-0" (192") – Door at east end – West to East

Plates

PIECE	SIZE	LENGTH	NOTES
Bottom plate	2x6	16'-0"	Cut out door opening after wall is stood up
Lower top plate	2x6	16'-0"	
Upper top plate	2x6	15'-1"	Shortened 5-1/2" each end (end truss overlap)

Stud Layout

#	POSITION	TYPE	LENGTH
1	0 - 1-1/2"	Stud	6'-10-3/4"
2	15-1/4 - 16-3/4"	Stud	6'-10-3/4"
3	31-1/4 - 32-3/4"	Stud	6'-10-3/4"
4	47-1/4 - 48-3/4"	Stud	6'-10-3/4"
5	63-1/4 - 64-3/4"	Stud	6'-10-3/4"
6	79-1/4 - 80-3/4"	Stud	6'-10-3/4"
7	95-1/4 - 96-3/4"	Stud	6'-10-3/4"
8	111-1/4 - 112-3/4"	Stud	6'-10-3/4"
9	127-1/4 - 128-3/4"	Stud	6'-10-3/4"
10	142-1/2 - 144"	King stud	6'-10-3/4"
11	144 - 145-1/2"	Jack stud	6'-5-1/4"
DOOR R.O. – 145-1/2" to 178-1/2" (33" wide)			
12	178-1/2 - 180"	Jack stud	6'-5-1/4"
13	180 - 181-1/2"	King stud	6'-10-3/4"
14	190-1/2 - 192"	Stud	6'-10-3/4"

Header

PIECE	SIZE	LENGTH	NOTES
Door header	(2) 2x6 + 1/2" ply spacer	3'-0"	Jack-to-jack outer (36")

- Cut list:
- 11 x 2x6 @ 6'-10-3/4" (82.75") – full-height studs + kings
 - 2 x 2x6 @ 6'-5-1/4" (77.25") – jack studs
 - 2 x 2x6 @ 3'-0" (36") – header plies
 - 1 x 1/2" plywood @ 5-1/2" x 36" – header spacer

West Wall 11'-1" (133") – South to North

Plates

PIECE	SIZE	LENGTH	NOTES
Bottom plate	2x6	11'-1"	
Top plate (single)	2x6	11'-1"	End truss bottom chord acts as 2nd plate

Stud Layout (10 studs)

#	POSITION	TYPE	LENGTH
1	0 - 1-1/2"	Stud	6'-10-3/4"
2	15-1/4 - 16-3/4"	Stud	6'-10-3/4"
3	31-1/4 - 32-3/4"	Stud	6'-10-3/4"
4	47-1/4 - 48-3/4"	Stud	6'-10-3/4"
5	63-1/4 - 64-3/4"	Stud	6'-10-3/4"
6	79-1/4 - 80-3/4"	Stud	6'-10-3/4"
7	95-1/4 - 96-3/4"	Stud	6'-10-3/4"
8	111-1/4 - 112-3/4"	Stud	6'-10-3/4"
9	127-1/4 - 128-3/4"	Stud *	6'-10-3/4"
10	131-1/2 - 133"	Stud	6'-10-3/4"

All 10 studs identical: Cut 10 x 2x6 @ 6'-10-3/4" (82.75")

* Stud #9: Leave OUT during assembly. Toenail in after wall is standing – no room for nailgun between last two studs.

East Wall 11'-1" (133") – South to North

Plates

PIECE	SIZE	LENGTH	NOTES
Bottom plate	2x6	11'-1"	
Top plate (single)	2x6	11'-1"	End truss bottom chord acts as 2nd plate

Stud Layout (10 studs)

#	POSITION	TYPE	LENGTH
1	0 - 1-1/2"	Stud	6'-10-3/4"
2	15-1/4 - 16-3/4"	Stud	6'-10-3/4"
3	31-1/4 - 32-3/4"	Stud	6'-10-3/4"
4	47-1/4 - 48-3/4"	Stud	6'-10-3/4"
5	63-1/4 - 64-3/4"	Stud	6'-10-3/4"
6	79-1/4 - 80-3/4"	Stud	6'-10-3/4"
7	95-1/4 - 96-3/4"	Stud	6'-10-3/4"
8	111-1/4 - 112-3/4"	Stud	6'-10-3/4"
9	127-1/4 - 128-3/4"	Stud *	6'-10-3/4"
10	131-1/2 - 133"	Stud	6'-10-3/4"

All 10 studs identical: Cut 10 x 2x6 @ 6'-10-3/4" (82.75")

* Stud #9: Leave OUT during assembly. Toenail in after wall is standing – no room for nailgun between last two studs.

Lumber Summary — All 4 Walls

Plates

QTY	SIZE	LENGTH
4	2x6	16'-0"
2	2x6	15'-1"
4	2x6	11'-1"

Studs

QTY	SIZE	LENGTH
45	2x6	6'-10-3/4"
2	2x6	6'-5-1/4"

Headers

QTY	SIZE	LENGTH
1	(2) 2x6 + 1/2" ply	3'-0"

Stock Lengths to Buy

QTY	STOCK
48	2x6 x 8'
4	2x6 x 12'
6	2x6 x 16'

Total: ~465 board-feet

Notes

Stud height = wall height (87.25") minus 3 plates (4.5") = 82.75" = 6'-10-3/4"
Jack stud height = door R.O. height = 77.25" = 6'-5-1/4"
N/S walls: lower top plate full-length, upper shortened 5-1/2" each end (end truss overlaps)
E/W walls: single top plate only (end truss bottom chord acts as 2nd plate)
E/W walls are shortened to fit between N/S walls (144" - 2 x 5.5" = 133")